VocaGrande Data Model

Design analysis and build plan, as of 2 Nov 2026

Philosophy

Vocabulary is the language: the more words you know, the more fluent you are perceived to be. The app counts **unique words (lexemes)** so its numbers are comparable with a dictionary's headword count (about 170,000 for English), while a separate count of **inflected forms** captures the ballooning, fluency-perceiving total. Goal: about 10,000 words each in Spanish and Italian by the end of 2027, with Merlin Diet menu vocabulary as a tracked slice.

Terminology

Term	Meaning
Lexeme / entry	The abstract dictionary word, identified by lemma + part of speech. The unit that is counted.
Lemma	The citation form you look up: *hablar*, *speak*, *casa*.
Form (inflected form)	*hablo, habla, hablas*; *speak, speaks, spoke*. A verb's full set is its paradigm.
Homograph	Different entries sharing a spelling (*lead* verb vs. metal). A relationship between entries, not a thing counted.
Polysemy / Sense	One entry with several related meanings. Senses sit inside an entry and are not counted toward the word total.
Part of speech (POS)	Noun, verb, and so on. Not the same as syntactic function (subject, object).
Syncretism	One spelling filling several paradigm cells: *habla* is present 3rd singular and imperative 2nd singular.

What is counted

- **Count:** unique (lemma, language, part of speech) entries, plus a

homograph index for same-POS homographs.

- **Do not count:** senses (translations and meanings) or inflected forms. Forms get their own separate fluency count.
- Gender is a property of the Lexeme, not the Form, for nouns. *casa* is feminine in singular and plural. Adjectives carry gender on Form.

Current model (built)

Lexeme field	Notes
`lemma`	Unique across the store, lowercase.
`englishMeaning`	Optional string. Stand-in for a future Sense.
`languageCode`	"es" for now.
`gender`	Optional `Gender` (masculine, feminine). nil means "na" (not a noun or not applicable). Added 2 Nov.
`notecardCorrectCount`, `notecardIncorrectCount`	Quiz tallies, in either direction.
`selections`	Captures (`TextSelectionItem`) this word was parsed from.

Known conflict: `lemma` is unique on its own, which will block (*lead*, verb) and (*lead*, noun) from coexisting. It must change to a compound key before part of speech is added.

Target model

```
Lexeme   lemma, languageCode, partOfSpeech, homographIndex,
gender?,
         pronunciation?, notecard counts
 |- Sense[]   gloss (English description), order, translations
-> [Lexeme]
 |- Form[]    text, mood?, tense?, person?, number?, gender?,
isIrregular
 '- VerbInfo? conjugation class (-ar/-er/-ir), stem-change
pattern, flags
```

```
ConjugationRule       class, mood, tense, person, number ->
ending, stem transform
ConjugationException  lexeme, mood, tense, person, number ->
text (flat overrides)

LearnerProfile        nativeLanguageCode, studyLanguageCodes,
goal, goalDate
```

- **Translation is a Lexeme-to-Lexeme relation held on Sense**, not on Lexeme. Meaning is what links words, so *lead* (to guide) maps to *guiar* and *lead* (to be ahead) to *liderar*. English words are first-class Lexemes, so quizzes, counts and forms work on both sides, and Spanish to Italian links come through shared English Senses.
- **Verb paradigms are feature tags on Form** (mood, tense, person, number as optional enums). The 501 Verbs grid is a view over a Lexeme's forms, not a new entity.
- **Forms are generated** by rules, then stem-change patterns, then exceptions override cell by cell. No bulk ChatGPT parse for forms.
- **No VocabWord entity.** It would add a layer without adding information. Grouping by spelling can be a computed view.
- **Language is a code string** on Lexeme (cheap to repeat), later growing into regions and dialects (es-MX, it-IT) if needed.

Decisions already made

- Count lexemes (lemma + POS), not forms and not senses.
- Role (POS) lives on Lexeme. Homograph is a query, with `homographIndex` for same-POS splits.
- Gender lives on Lexeme, with "na" meaning no gender. The ChatGPT check verifies the article (el, la, na) along with the translation.
- Keep Form feature fields optional from the start so later stages need no migration.
- Persist only the candidates the learner picks, not whole lookup results.
- A word is "known" by a rule over the notecard counts (for example, 3 correct and above 80%). The 10,000 goal counts known words.

Build plan

Stage	Content	Status
1	Lexeme by spelling, translation, notecard quiz	Done
1a	iPhone list and add screen; `gender` and el/la/na with ChatGPT check	Done
2	Add `partOfSpeech`, `homographIndex`, compound uniqueness; parse returns POS	Next
3	Add Sense; ? button asks ChatGPT for candidate meanings; pick one or import all; quiz accepts any valid translation	Planned
4	Form with text only, recorded by the parse	Planned
5	Feature tags on Form (mood, tense, person, number)	Planned
6	VerbInfo and the rules engine for regular verbs	Planned
7	Stem-change rules, then ConjugationException; 501 Verbs as the irregular test set	Planned
8	Paradigm view and richer quizzes	Planned
Later	LearnerProfile, Italian, language picker in the quiz, topic tag (for example "food")	Planned

Open questions

- Exact rule for a "known" word, and whether the learned count should require both directions.
- Whether import-all from a lookup should raise the total or only add candidates to a separate pool.
- How to migrate existing `englishMeaning` strings into Senses (plan: one Sense per Lexeme).
- How to handle existing data when POS is added (re-parse or default to "unknown").